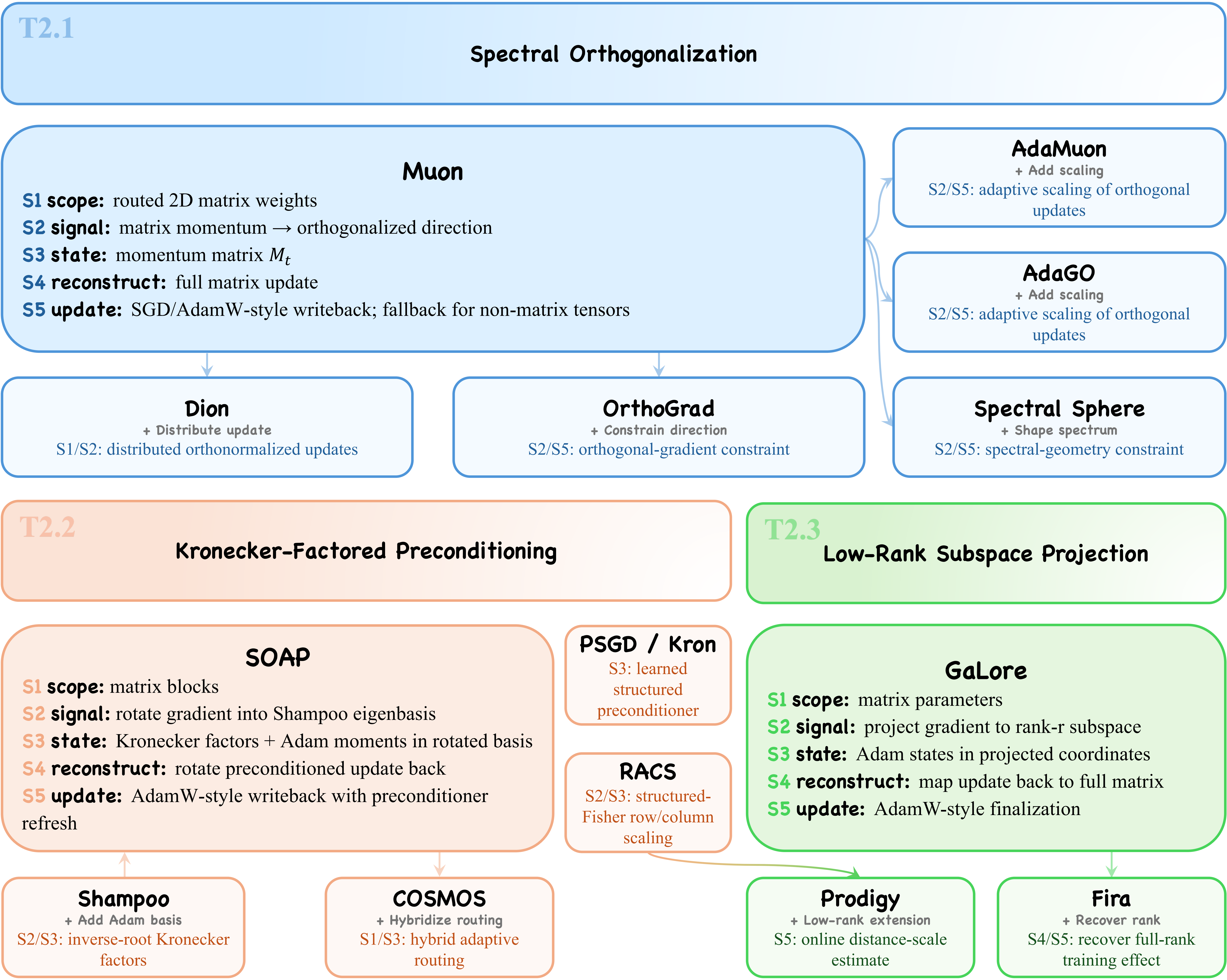

T2.1
Spectral Orthogonalization
Muon
S1 scope: routed 2D matrix weights
S2 signal: matrix momentum → orthogonalized direction
S3 state: momentum matrix M_t
S4 reconstruct: full matrix update
S5 update: SGD/AdamW-style writeback; fallback for non-matrix tensors
AdaMuon
+ Add scaling
S2/S5: adaptive scaling of orthogonal updates
AdaGO
+ Add scaling
S2/S5: adaptive scaling of orthogonal updates
Dion
+ Distribute update
S1/S2: distributed orthonormalized updates
OrthoGrad
+ Constrain direction
S2/S5: orthogonal-gradient constraint
Spectral Sphere
+ Shape spectrum
S2/S5: spectral-geometry constraint
T2.2
Kronecker-Factored Preconditioning
T2.3
Low-Rank Subspace Projection
SOAP
S1 scope: matrix blocks
S2 signal: rotate gradient into Shampoo eigenbasis
S3 state: Kronecker factors + Adam moments in rotated basis
S4 reconstruct: rotate preconditioned update back
S5 update: AdamW-style writeback with preconditioner refresh
PSGD / Kron
S3: learned structured preconditioner
RACS
S2/S3: structured-Fisher row/column scaling
GaLore
S1 scope: matrix parameters
S2 signal: project gradient to rank-r subspace
S3 state: Adam states in projected coordinates
S4 reconstruct: map update back to full matrix
S5 update: AdamW-style finalization
Shampoo
+ Add Adam basis
S2/S3: inverse-root Kronecker factors
COSMOS
+ Hybridize routing
S1/S3: hybrid adaptive routing
Prodigy
+ Low-rank extension
S5: online distance-scale estimate
Fira
+ Recover rank
S4/S5: recover full-rank training effect